

Keelan O'Doherty

Ann Arbor, MI | 602.376.2533 | keelan.odoherty@icloud.com | linkedin.com/in/keelanodoherty | github.com/Keelan4604 | U.S. Citizen, ITAR Eligible

EDUCATION

University of Michigan <i>Master of Science (MS) in Aerospace Engineering, Space Systems</i> Relevant Coursework: Spacecraft Technology, Space Policy & Management, Linear Systems Theory, Management of Space Systems Design, Astrodynamics	Ann Arbor, MI · Aug 2026 – Dec 2027
University of Alabama, Honors College <i>Bachelor of Science (BS) in Aerospace Engineering, Minor in Mathematics</i> - Honors GPA 3.64/4.0 · Dean's List: every semester, Fall 2022 – Spring 2025 - Scholarships: Full-Ride Academic, Presidential, Engineering, Taylor Lewan Offensive Line Scholarship	Tuscaloosa, AL · Aug 2022 – May 2026
Florence University of the Arts, Study Abroad	Florence, Italy · May 2024

ENGINEERING EXPERIENCE

Northrop Grumman Sponsored Lunar Cargo Module (CAT-LIT) <i>Systems Engineer, Multidisciplinary Design Program</i> - Systems engineer on a 7-person team designing a standardized cargo module to land 8,000 kg on the lunar surface, compatible with Starship and Blue Moon - Developing the Level 1–4 requirements set and per-requirement verification methods (NASA SP-2016-6105); maintaining the vehicle mass and power budgets	Ann Arbor, MI · Sep 2026 – Present
University of Alabama Aerodynamics Research Laboratory (AAERO) <i>Undergraduate Research Assistant, Vertical Flight Society Chapter</i> - Designed, SLA 3D-printed, and bench-tested 4 UAV propeller geometries at 10,000 RPM to characterize thrust against acoustic performance - Built the test setup and authored run procedures using load-cell thrust instrumentation and microphone acoustic profiling, structured for repeatable comparison across geometries - Correlated measured results against Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA) predictions to identify model-to-hardware divergence; presented findings at the URCA Conference 2025	Tuscaloosa, AL · Dec 2024 – May 2026
Fixed-Wing eVTOL Aircraft - Design, Build & Flight Test <i>Senior Capstone</i> - Modeled the full ORCA airframe in SolidWorks from scratch for 3D printing, adapted from an existing VTOL design (fuselage, wings, spars, control surfaces, motor mounts), including a modular payload nose for optical, IR, and thermal sensor swaps - Served as team pilot and ran the flight test campaign: phased hover and vertical-to-forward transition trials validating stability, control response, and endurance - Modeled propulsion tilt-path dynamics and servo actuation sequencing in MATLAB/Simulink; validated the power budget against measured draw	Aug 2025 – May 2026
Wilson & Company <i>Engineering Intern</i> - Gained hands-on experience with construction materials and techniques through 50+ hours of on-site exposure, documenting nonconformances and as-built conditions - Collaborated with senior engineers to develop 5+ project plans and timelines, and contributed to preparation of 5+ reports for client presentations and regulatory submissions	Phoenix, AZ · Jul 2024 – Aug 2024

LEADERSHIP & ADDITIONAL EXPERIENCE

Pi Kappa Alpha Fraternity <i>Class President; Sports Chair</i> Elected by peers to lead a pledge class and to organize and schedule intramural athletics across a 100+ member chapter	Tuscaloosa, AL · 2022 – 2026
Watercraft Restoration - Owner & Technician <i>Phoenix, AZ</i> - Restored 65+ watercraft end to end, including 30+ full engine rebuilds and 30+ complete electrical rewires, from teardown through functional test - Fabricated wire harnesses and mechanical subassemblies to torque specification; wrote repeatable teardown, build, and checkout procedures that reduced rework on later units	Feb 2021 – Aug 2026
PowerPro Rentals LLC - Founder <i>Phoenix, AZ</i> Founded and operated a profitable heavy equipment rental LLC, handling market research, pricing, purchasing, supply chain, logistics, and finances; built partnerships with 10+ local businesses	May 2023 – Aug 2023

TECHNICAL PROJECTS

- Embedded Vehicle Control System** - built Arduino-based control with GSM/IoT telemetry (SIM7000A) over LTE CAT-M1 for remote monitoring and actuation; designed power distribution using transistor logic and relay control, and wired the physical harness
- Vehicle Systems Diagnostics** - electrical and mechanical fault isolation, rewiring, and CANbus analysis on automotive platforms

TECHNICAL SKILLS

Build & Test: Mechanical and engine assembly, wire harness fabrication, electrical troubleshooting (multimeter, continuity, compression), load-cell instrumentation, SLA and FDM 3D printing, torque and fastener practice, test procedure development, Circuit Design

Design & Analysis: SolidWorks, AutoCAD, Computational Fluid Dynamics (CFD), ANSYS Fluent, SolidWorks Flow Simulation, Finite Element Analysis (FEA), MATLAB/Simulink, Data Analysis

Software: Python, Arduino/C++, Linux, Docker, Git/GitHub, Microsoft 365, Google Workspace

Additional: Languages: Spanish · Musical: Piano, Cello, Guitar